PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 5

How sure are we?

Describing data, propagating uncertainty
and testing hypotheses

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Where we left off

From Lectures 3 and 4

- Fitted parameters come with uncertainties and correlations
- χ^2/ν and residuals judge a fit
- Independent errors add in quadrature; offsets add linearly
- Unmodelled time variability is not noise

Today

- Describing data: location, spread, distributions
- Propagating uncertainty into computed quantities
- Testing hypotheses: is a difference real?
- One running example: measuring g with a pendulum

In every step the key question will be the same: are the values independent?

Three questions about any result

Task	Question	Tools	Part
Describe	What do the data look like?	Plot in order; median, quantiles; bootstrap; a distribution family only when known	A
Propagate	How uncertain is a quantity computed from them?	Monte Carlo; the linear rule as a shortcut	B
Test	Is an observed difference real, or chance?	Permutation and rank tests; parametric tests when the distribution is known	C

The order matters

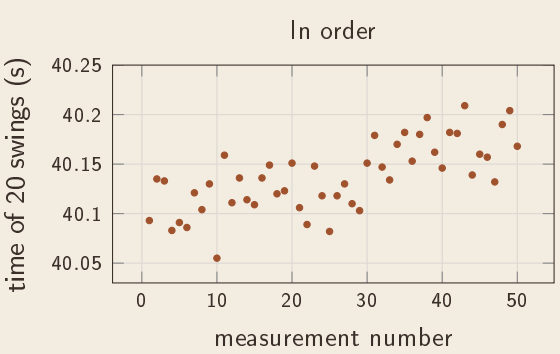
A wrong description (e.g. missed drift or a bad value) gives a wrong uncertainty in Part B and a wrong test in Part C.

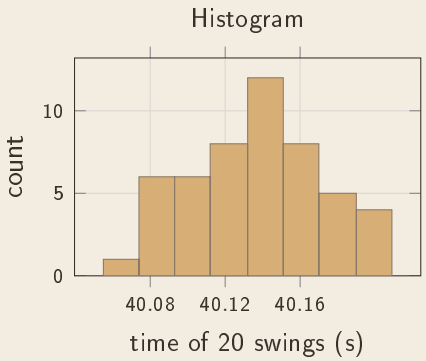
PART A

Describing data

What do the values look like – in order, and as a distribution?

Plot the values in the order they were taken





The histogram looks like ordinary scatter. In order, the times drift upwards: the first 25 average 40.115 s, the last 25 average 40.159 s. Successive values are not **independent** (independent: knowing one value tells nothing about the next).

Location and spread

Values $x_1, \ldots, x_N$; here a new set of $N = 9$ timings of 20 swings: 40.12, 40.18, 40.09, 40.15, 40.11, 40.21, 40.13, 40.16, 40.10 s. **Median**: the middle value when sorted (even N: the mean of the two middle ones).

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}.$$

median	40.130 s
mean $\bar{x}$	40.139 s
standard deviation s	0.040 s

s describes the spread of **single** measurements: here 6 of the 9 timings lie within $\bar{x} \pm s$.

Why $N - 1$?

Deviations are measured from $\bar{x}$, fitted to the same data, so one value is used up: $\nu = N - 1$ degrees of freedom (as for χ^2, Lecture 3).

One bad value: mean versus median

	9 timings	+ one value of 41.14 s
mean	40.139	40.239
standard deviation s	0.040	0.319
mean absolute deviation (from the mean)	0.032	0.180
median	40.130	40.140
MAD: **median** absolute deviation	0.030	0.035

MAD: the median of the distances $|x_i - \text{median}|$. Half of the values lie within $\pm$MAD of the median, whatever the distribution. Each spread changes by itself: s eightfold, the mean absolute deviation almost sixfold, the MAD from 0.030 to 0.035 s.

Never delete a value silently

Find the cause (here: the watch stopped one second late), and report what you did. A value that is merely surprising may be the physics.

Nonparametric and parametric descriptions

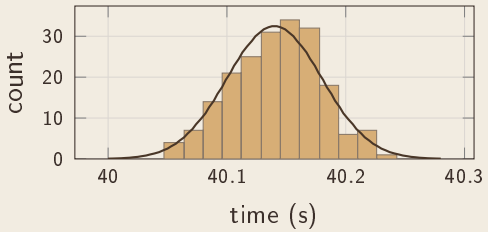
	Nonparametric	**Parametric**
Assumes	no shape; uses order, ranks, quantiles*	a family (Gaussian, Poisson)
Centre and spread	median, MAD; also $\bar{x}$, s (no shape implied)	$\bar{x}$, s as estimates of μ, σ
Strength	any shape; median, MAD robust to bad values	analytical expressions for uncertainties and p-values
Risk	none from the shape (no shape is assumed)	wrong if the shape is wrong; few values cannot check it

*rank: position in the sorted list; quantile: the value below which a given fraction lies (median: 50%).

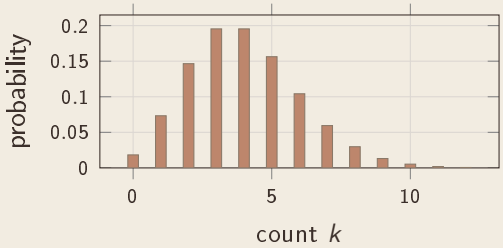
Use a distribution family only when prior physical knowledge supplies it (next slide).

When the distribution is known: Gaussian and Poisson

200 simulated timings; Gaussian curve



Poisson: mean $\mu = 4$



Gaussian arises when many small independent effects add up. 68.3%, 95.4%, 99.7% of values lie within $\pm 1, 2, 3$ standard deviations of the mean.

Poisson counts of independent events, $p(k) = \mu^k e^{-\mu}/k!$; standard deviation $\sqrt{\mu}$ (the counting data of Lecture 3).

Use only when physics or an independent calibration supplies the family; check it against the data (histogram, plot in order).

Standard deviation and standard error

Given $N = 9$ measured timings of 20 swings (slide 6), how close is their mean $\bar{x}$ to the true, but unknown, value x^*?

The uncertainty of $\bar{x}$ is expressed by the **standard error of the mean**, which reads

$$\boxed{s_{\bar{x}} = \frac{s}{\sqrt{N}}} = \frac{0.040}{\sqrt{9}} = 0.013 \text{ s},$$

where s is the standard deviation of the individual timings (slide 6). $s_{\bar{x}}$ is the standard deviation that $\bar{x}$ would have over repeated sets of N measurements.

If we took N timings with the same $s = 0.04$ s:

N	10	100	1000
s	0.04	0.04	0.04
$s/\sqrt{N}$	0.013	0.004	0.0013

Condition

The expression for $s_{\bar{x}}$ relies on the assumption that the measured values are independent. If successive values are related (e.g. the drift of slide 5), it does not give the uncertainty of the mean.

Quoting a result

Problem: how do we write the period P of one swing from the 9 timings of 20 swings?

$P = \bar{x}/20 = 40.139/20 = 2.0069\,\text{s}$. Dividing by 20 also divides the standard error by 20: $0.013/20 = 0.0007\,\text{s}$ (slide 10).

Write

$P = 2.0069 \pm 0.0007\,\text{s}$
($\pm$ one standard error of the mean, $N = 9$)

Not

$P = 2.006944 \pm 0.000660\,\text{s}$

Rules: give the uncertainty to 1–2 significant figures, because it is itself uncertain; round the value to the same decimal place; give the units; say what $\pm$ means and how it was obtained.

The next slide gives the uncertainty without a formula; Part B shows results whose uncertainty is asymmetric, quoted as an interval.

The bootstrap: uncertainty without assuming a distribution

Problem: from N measured values $x_1, \ldots, x_N$ we compute a statistic, e.g. the median. How uncertain is it? We do not assume any form of the distribution.

Idea: use the N values themselves as the best available picture of how the values vary. No new data is created.

1. Draw N values at random from the N; each can be drawn again.
2. Compute the statistic from them.
3. Repeat this B times. The standard deviation of the B results is the standard error of the statistic.

Example: the 9 timings of slide 6, with $B = 20\,000$ repetitions: median $= 40.130 \pm 0.019$ s.

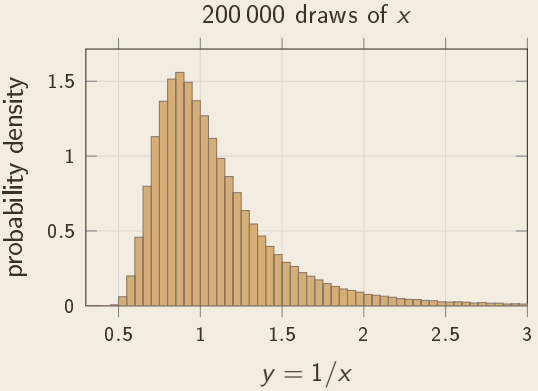
Condition: independent values.

PART B

Propagating uncertainty

From the uncertainty of what we measure to the uncertainty of what we compute.

Propagating through any function: Monte Carlo



Problem: $y = f(x)$ with x uncertain. How uncertain is y? Example: $y = 1/x$, with x a normally distributed random variable with mean 1.0 and standard deviation 0.3.

Monte Carlo: draw many values of x, compute y for each, and describe the y values.

Inputs: bootstrap samples of the measured x; or, if x is known only by a stated tolerance, draws from that.

Result: median 1.00; the middle 68% of the y values lie between 0.77 and 1.43, an **asymmetric** uncertainty.

When f is nearly straight: the linear rule

Problem: $y = f(x)$, where x has standard error σ_x. Can a formula replace the simulation?

If f is nearly straight over $\bar{x} \pm \sigma_x$:

$$\boxed{\sigma_y \approx |f'(\bar{x})|\,\sigma_x}$$

Check against Monte Carlo (slide 14), for $y = 1/x$ at $\bar{x} = 1$:

σ_x	linear rule	Monte Carlo, middle 68%
0.05	0.95–1.05	0.95–1.05
0.3	0.70–1.30	0.77–1.43

When the function is nearly straight, the rule works. When it is curved, the rule fails: use Monte Carlo.

Several independent inputs: the pendulum

For $y = f(x_1, \ldots, x_n)$ with **independent** inputs x_i of standard uncertainties σ_i:

$$\sigma_y^2 = \sum_i \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_i^2$$

for $g = \dfrac{4\pi^2 L}{P^2}$: $\left(\dfrac{\sigma_g}{g}\right)^2 = \left(\dfrac{\sigma_L}{L}\right)^2 + \left(\dfrac{2\sigma_P}{P}\right)^2$

	L	P	total (in quadrature)
value	1.000 ± 0.002 m	2.0069 ± 0.0007 s	
contribution to σ_g/g	0.20%	$2 \times 0.033\% = 0.07\%$	0.21%

Result: $g = 9.80 \pm 0.02\ \mathrm{m\,s^{-2}}$

Check by Monte Carlo (slide 14): take P from 20 000 bootstrap samples of the mean of the 9 timings (slide 12), divided by 20, and L from its stated tolerance (Gaussian, ± 0.002 m, the only information we have about the ruler). Compute g for each: standard deviation $0.020\ \mathrm{m\,s^{-2}}$. The linear rule agrees because g is nearly straight over these small ranges.

To improve the result, reduce the **largest** contribution first: here L.

Random and systematic errors

Problem: every measurement error has a part that changes from one measurement to the next, and a part that stays the same. How does each part affect the result?

	Random	**Systematic**
What it is	differs from one measurement to the next (independent)	the same in every measurement
Example	reaction time when stopping the watch	a ruler that is 0.1% too long
Taking more measurements	averages out, as $s/\sqrt{N}$	does not average out
Seen in s, $s/\sqrt{N}$, the bootstrap	yes	no: only by an independent check (another instrument, a calibration)

Message: statistics computed from the data measure only the random part. A systematic error must be estimated separately and quoted alongside.

PART C

Testing hypotheses

Is a difference real, or could chance have produced it?

The p-value

Problem: we have two sets of measured values. Do they come from the same distribution, or are the two distributions different?

- **Null hypothesis H_0:** both sets come from the same distribution, so any difference between them is due to chance.
- **Test statistic:** a number that measures how far apart the two sets are, e.g. the difference of their medians.
- **p-value:** the probability, if H_0 is true, of a test statistic at least as large as the one observed. $0 \leq p \leq 1$.

Small p: if H_0 were true, a difference at least this large would occur only rarely (e.g. $p = 0.01$: in 1% of cases). This is evidence that the two distributions differ, or that a test assumption fails.

Large p: if H_0 were true, a difference at least this large would occur often (e.g. $p = 0.4$: in 40% of cases). This does not show that the distributions are the same. **To claim they are practically the same**, show that the interval for their difference lies within a tolerance chosen beforehand.

Two families of tests

To compute p we need a **reference distribution**: how the test statistic would be distributed if H_0 were true.

	Nonparametric	Parametric
Reference distribution	built from the data themselves: by reshuffling, or from ranks	derived from an assumed distribution of the data
Examples	permutation tests, rank tests, shuffle surrogates	Student's t, χ^2
Assumes	independence only	independence and the distribution
Valid when	any shape	the assumed distribution is right (or N is large)

Common to both: The definition of p is the same; only the way its reference distribution is obtained differs. Neither family protects against dependent values or systematic errors.

Do two groups differ? Permutation and rank tests

Problem: do groups A and B differ? H_0: same distribution.
A: the 9 timings of slide 6 + the bad value 41.14 s.
B (second session): 40.17, 40.26, 40.20, 40.23, 40.19, 40.28, 40.22, 40.24, 40.18, 40.21 s.

Permutation: under H_0 the labels do not matter. Shuffle them many times, recompute the difference of medians; p = fraction at least as large as observed.

Rank (Mann–Whitney): rank all 20 values; are A's ranks unusually low or high?

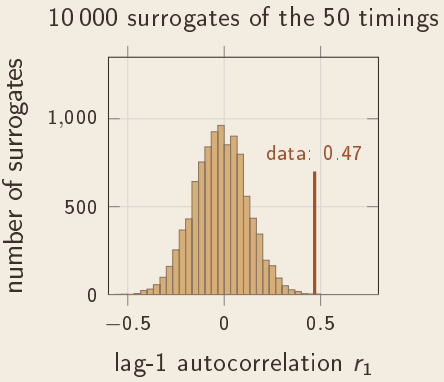
p (two-sided)	with bad value	without
permutation	0.005	0.002
rank	0.011	0.002
t on means*	0.84	0.0003

*parametric t-test: assumes Gaussian timings

Permutation and rank tests see the difference with or without the bad value; the t-test loses it.

Size of the difference: the median of all 100 differences B − A (Hodges–Lehmann): 0.07 s, 95% interval 0.03–0.11 s; without the bad value 0.08 s, 0.04–0.12 s.

Are successive values independent? Shuffle test



Problem: are successive values of a series related, or independent?

Idea: if the values are independent, their order does not matter. Shuffling keeps all the values (the same histogram) but destroys any order. The shuffled copies are **surrogate data**: copies made to obey H_0 while keeping everything else.

Lag-1 autocorrelation r_1: the correlation between each value and the next, $r_1 = \sum_i (x_i - \bar{x})(x_{i+1} - \bar{x}) / \sum_i (x_i - \bar{x})^2$. It is near 0 for independent values and near 1 if neighbours are alike.

Result: the 50 timings of slide 5: $r_1 = 0.47$. Only 2 of 10 000 surrogates reach it, so $p \approx 0.0002$: successive timings are not independent (the drift).

The assumptions behind every test

Assumption	**If it fails**
Values are independent	standard errors and p-values do not hold – common for time series; hence use time-series analysis methods
The distribution is right (parametric tests)	use a nonparametric test: permutations, ranks, surrogates
The test was chosen before looking	otherwise, among many tried tests some will give $p < 0.05$ by chance alone
No systematic errors	p says nothing about them

A p-value is only as good as its assumptions.

Summary

Task	Tools	Check
Describe	plot in order; median, MAD, quantiles; bootstrap; $\bar{x}$, s; a distribution family only when known	independence; bad values
Propagate	Monte Carlo; the linear rule when f is nearly straight	curvature of f; systematic errors
Test	permutations, ranks, surrogates; t, χ^2 when the distribution is known	assumptions; number of tests; report the effect with its interval

Before you report a result: five questions

1. Did I plot the values in the order they were taken? *(drift, time variability)*
2. Are the values independent? *(otherwise uncertainties and p-values do not hold)*
3. Which uncertainties are random, and which systematic? *(random ones average out with more data; systematic ones do not, and must be estimated separately)*
4. Does my method assume a distribution or a straight-line approximation, and do I know that from prior physical knowledge? *(if not: resampling, permutations, ranks, Monte Carlo)*
5. What exactly is H_0, and did I choose the test before looking? *(many tests: some 'significant' by chance)*

Then quote: value, interval (or $\pm$), units, and how they were obtained.

TO TAKE AWAY

A number without an uncertainty is not a result.

And a p-value without its assumptions is not evidence. Plot in order, check independence, propagate honestly, then test.